BIPI

Ghana's Ecological Pulse

"Transforming climate awareness into climate action through technology"

---

1. The Name: Why BIPI

Ghanaians do not say 'blood pressure' in everyday conversation. They say BP. When someone is stressed, you hear it: 'My BP is too high.' When there is a health emergency, the first thing a doctor wants to know is your BP. Everyone, across every background, understands what that means. It is a signal. It tells you something is wrong underneath the surface before the full crisis arrives.

That is exactly what this platform does, but for Ghana's environment and the people who live in it. BIPI takes the ecological blood pressure of the country. It reads the warning signs in how people understand, respond to, and prepare for climate change. It tells us where knowledge is low, where communities are most at risk, and where action is needed before the floods arrive, before the harvest fails, before the next disaster catches people unprepared.

The name is not just branding. It is the thesis of the entire project. Just as a doctor cannot treat what they cannot measure, Ghana cannot respond to climate vulnerability it cannot see. BIPI makes it visible.

BIPI: Know the Signs. Act Before the Crisis.

---

2. The Problem We Are Solving

Ghana is climate-vulnerable. And it is under-informed about that vulnerability.

Ghana faces serious, documented climate risks. Accra floods almost every major rainy season, and the damage falls hardest on communities in low-lying settlements with poor drainage. The northern regions experience erratic rainfall, prolonged dry spells, and food insecurity that worsens year by year. Keta and communities along the coast are literally losing land to rising sea levels and coastal erosion. Temperatures are rising, and heat stress is becoming a real occupational and health hazard for outdoor workers, pregnant women, and people with chronic illness.

These are not abstract projections. They are things happening to Ghanaians right now. And the people most exposed to these risks are also the people with the least access to information that could help them respond.

The four gaps that BIPI is designed to close:

Gap 1: No accessible, local-language climate education platform exists in Ghana.

The tools that do exist assume that users have smartphones, reliable internet, English literacy, and free time to sit and read long articles. None of those assumptions hold for most Ghanaians. A woman farmer in the Upper East does not have time to browse a website. A young man in Nima with a low-cost Android phone and a patchy data connection cannot stream videos. A person with a visual impairment cannot use a platform with no screen-reader support. The result is that the people who need climate information most are the last to receive it.

Gap 2: Women bear the highest climate burden but receive the least targeted support.

Women in Ghana are disproportionately affected by climate change because of how their responsibilities are structured. When water becomes scarce, it is women who travel further to collect it. When harvests fail, it is women managing household food budgets who absorb the shock. When disasters strike, women face specific safety risks in shelters and evacuation situations that are rarely planned for. Despite this, virtually no digital climate education tool has been designed with women's lives, languages, and daily realities in mind. BIPI addresses this directly with a dedicated track, audio-first content in Twi and Pidgin, and scenarios built around situations women actually face.

Gap 3: Persons with disabilities are almost entirely excluded.

Climate education platforms in Ghana, to the extent that they exist at all, are built for a default user: sighted, literate, connected, and able-bodied. Persons with visual impairments, hearing impairments, mobility challenges, or cognitive differences are not considered in the design. This is not just a technical failure. It is an equity failure. People with disabilities face heightened climate vulnerability, particularly during disasters when evacuation plans rarely account for mobility or communication needs. The GreenRes curriculum matrix dedicates eight full modules (50 through 57) to disability inclusion. BIPI builds those modules into the platform from day one, not as an afterthought.

Gap 4: Government agencies cannot see where climate knowledge is failing.

NADMO prepares for disasters. EPA Ghana tracks environmental change. MoFA advises farmers. But none of these agencies has access to a real-time picture of where climate knowledge gaps are most severe, which communities are most poorly informed, or which demographics are falling furthest behind. They deploy resources and run campaigns without the data to know where those resources will have the most impact. BIPI generates that data as a natural byproduct of people using the platform to learn.

Ghana has no map of its own climate knowledge gaps. BIPI builds that map, passively, in real time, from the ground up.

---

3. The Solution: What BIPI Is

BIPI is a Progressive Web App, which means it is a website that behaves like a mobile application. Users install it directly from a browser link, with no app store involved, no lengthy download, and no storage-heavy install process. Once installed, it works fully offline, caching all lessons, audio, and user progress on the device itself. It runs on the same low-cost Android phones that most Ghanaians already carry.

The platform delivers all 120 modules of the GreenRes curriculum through five distinct content formats, so that every user, regardless of literacy level, language, device capability, or physical ability, can engage with the material in a way that actually works for them.

Underneath the learning experience, every user interaction is silently tagged with the user's region, demographic category, the topic they engaged with, and their outcome. This data is anonymised, aggregated, and fed into a live national intelligence dashboard called the BIPI Pulse, which is Ghana's first citizen science climate knowledge map, built passively from the ground up by ordinary people going about their learning.

The core insight driving the whole design is this:

Users just use the app. The national intelligence builds itself.

---

7. BIPI Pulse: The Citizen Science Intelligence Layer

BIPI Pulse is the feature that makes this platform genuinely different from every other climate education app. It is directly aligned with Module 97 of the GreenRes curriculum, which is titled Citizen Science and which calls for exactly this: a citizen science dashboard, interactive reporting map, community leaderboard, and mobile reporting interface. BIPI Pulse is the fulfilment of that module, not an addition to it.

Here is the underlying logic. When thousands of Ghanaians use BIPI to learn about climate change, their behaviour as learners generates data. What topics do people in Tamale fail most often? Where in the Western Region do users drop off before completing the disaster preparedness module? Which women's modules are most completed? Which disability modules have the lowest engagement? None of this information exists in Ghana today. BIPI generates it passively, without asking users to do anything extra, simply by observing how they learn.

Data stream	What it becomes	Who it serves
Quiz answers by region and topic	Heat map showing which districts fail specific climate knowledge areas most frequently	NADMO, EPA Ghana, district disaster management authorities
Gender breakdown by module	Gender knowledge gap map showing where women's scores diverge most from men's on identical modules	NGOs, MoFA gender desk, women's development organisations
Disability user performance by module	Accessibility gap map showing which modules are failing users with disabilities, and in what ways	Disability rights organisations, Prof. Naami's research team, NCPD
Drop-off points per module	Content failure map showing exactly where users abandon a module, which signals where content needs to be redesigned or simplified	BIPI platform team and content reviewers
Completion rates by district	Engagement map showing which regions are most and least active, useful for targeting outreach and distribution partnerships	ACC, Mastercard Foundation, GES
Wrong answers by topic across all users this week	Trending knowledge gap report showing what Ghana is collectively misunderstanding about climate change right now	All institutional partners, media, policy teams

The BIPI Pulse dashboard is a separate, partner-facing interface. Regular users never see it. Institutional partners, including government agencies, NGOs, researchers, and development organisations, access it through a separate login. They see live heat maps, trend charts, demographic breakdowns, and module-level performance data. No individual user is ever identifiable in any output. Only aggregated, district-level data is displayed.

Users just learn. The national intelligence builds itself.

---

8. The Full User Pipeline: From First Open to National Intelligence

This section describes exactly what happens at every stage of a user's journey through BIPI, from the moment they first open the app to the moment their learning behaviour becomes part of Ghana's national climate intelligence picture.

Stage 1: First Open

The app opens with zero friction. There is no login screen. There is no registration form. There is no email field. The very first thing the user sees is a single question, displayed in Twi by default, asking them how they prefer to use the app. They pick their language. That is it. They are inside the platform in under thirty seconds without having given away a single piece of personal information.

Stage 2: Profile Setup (Three Taps, No Typing)

BIPI asks three questions at onboarding, all answered by tapping an icon. No keyboard appears.

- First tap: I am... Youth / Woman / Person with disability / Other. Icon-based. No reading required.
- Second tap: I care most about... Food / Water / Jobs / Safety / My community. User picks up to two.
- Third tap: I prefer to... Read / Listen / Watch. This sets the default content format for every module that follows.

These three answers are stored locally on the device and used to personalise every part of the experience from this point forward. The platform knows who it is talking to and adjusts accordingly.

Stage 3: Consent Screen (Ghana Data Protection Act 2012 Compliant)

Before the user enters the main platform, they see one screen. It is written in plain language and displayed in Twi, Pidgin, and English based on the language they picked. It says:

> **"BIPI learns from how people use it to help Ghana understand climate risks better. Your answers are never linked to your name or your phone. Do you agree to share your learning data in this anonymous way?"**

If the user taps YES, their anonymised interaction data contributes to BIPI Pulse.

If the user taps NO, they still access the full platform with no restriction. Their data simply does not enter the aggregation pipeline.

Stage 4: The Learning Experience

The platform serves the right content format based on what the user selected at onboarding. A woman who chose Listen receives audio stories delivered in Twi. She never sees a wall of text. A youth who chose Watch receives timed scenarios and news clips. A person with a visual impairment who chose Listen receives full audio with no visual dependency and screen-reader compatible navigation throughout. The content is the same across all formats. The delivery is tailored to the person.

Stage 5: Silent Data Capture

Every time a user answers a question, completes a module, replays a section, or abandons partway through, BIPI logs the following information in the background, invisible to the user: the module ID, the topic, the track, the content format used, the answer given, whether it was correct, and the time spent. This is tagged with the user's region, their demographic category from onboarding, and their language preference. It is never tagged with their name, phone number, or any identifying information.

Stage 6: Feedback and Reward

After completing a module or a section, the user receives immediate feedback. Their progress bar advances. They earn XP points. If they have completed enough modules in a track, a district badge unlocks: Accra Climate Defender, Volta Resilience Champion, Northern Preparedness Leader. Their streak counter updates. If they have completed a reflective module, their journal entry is saved privately. The leaderboard for their district updates. They feel the progress. They come back.

Stage 7: BIPI Pulse Updates

In the background, the anonymised data from their session is aggregated with data from all other users who have consented. The BIPI Pulse dashboard updates in near-real time. Heat maps shift. Trending knowledge gaps change. District rankings update. Institutional partners who log into the partner dashboard see a richer picture of Ghana's climate knowledge landscape after every session.

Stage 8: The Loop Closes

BIPI uses its own data to improve itself. Modules with consistently high drop-off rates are flagged for content review. Formats that underperform with specific demographics are reconsidered. Strong modules are promoted to new users first. The platform gets measurably better with every cohort of users who pass through it.

---

10. Technical Architecture and Deployment

The Application Stack

Every technology choice in BIPI's stack was made with two constraints in mind: it must be buildable by a five-person student team in three months, and it must be reliable in low-connectivity, low-device environments in Ghana. There are no experimental frameworks. No over-engineered dependencies. Just proven tools assembled deliberately.

Layer	Technology	Why this choice
Frontend framework	Next.js 15 with App Router, TypeScript,	Widely used, excellent PWA support, strong offline caching via service workers, works well

	Tailwind CSS	with Supabase. All team members can contribute to it.
PWA shell	Service Worker via next-pwa or Workbox	Handles offline caching of all module content, audio files, and user progress. Users with zero connectivity after first load still have full access to everything they have already encountered.
Backend and database	Supabase (Postgres, Auth, Storage)	Handles user auth, data storage, and real-time updates. Free tier is sufficient for the build and pilot phase. Scales cleanly. The team already knows it from the Project_Mimi architecture.
AI and content generation	Claude API (claude-sonnet-4-6) as primary, with a full offline fallback	Used to generate module content drafts from the curriculum matrix, localise terminology into Twi and Pidgin, and power the AI career advisor in the youth track. See Section 11 for the full fallback plan.
Maps and regional data	Mapbox GL JS or Leaflet.js with Ghana district boundary GeoJSON	Powers the regional heat maps in BIPI Pulse. The GIS background of one team member makes this achievable within the timeline.
Text-to-speech	Web Speech API (browser-native, no external dependency) as first pass	Every modern Android browser supports it. No cost, no API key, works offline for cached content. Upgradeable to a dedicated Ghanaian-language TTS model in a later phase.
SMS and USSD fallback	Africa's Talking API	Extends platform reach to users with basic phones and no data connection. Core quiz questions can be delivered by SMS. Responses can be received by SMS. This is the fallback for the most marginalised users.
Video content	Embedded YouTube links or direct MP4 links to publicly available Ghana news footage	No production cost. Real footage from Joy News, Citinewsroom, and GhanaWeb already exists for every major Ghana climate event. We curate, we do not produce.
Push notifications	Web Push API via service worker	Sends proactive module reminders, streak alerts, and new badge notifications without requiring the app to be open.

Deployment and Access

BIPI is deployed as a Progressive Web App at a public URL. Deployment uses Vercel (free tier for Next.js projects) for the frontend and Supabase cloud for the backend. The deployment process is:

- User receives a WhatsApp link, a QR code, or a direct URL
- They open the link in any modern mobile browser (Chrome on Android, Safari on iOS)
- The browser prompts them to add the app to their home screen
- One tap. The app icon appears. It launches like any installed app.
- On subsequent launches, the service worker loads all cached content instantly, even with no internet connection

There is no Play Store submission. No iOS App Store review. No 50MB download. No storage requirement beyond what the user's browser allocates. This is the correct deployment model for reaching users in Ghana's lower-income communities, where storage is scarce and data is expensive.

The Actors: Who Uses BIPI and How

Actor	How they access BIPI	What they do on the platform	What BIPI gives them
General learner (youth, adult)	Installs via browser link on Android phone	Completes modules, earns badges, tracks streak, competes on leaderboard	Climate knowledge, career pathway guidance, district badge, leaderboard ranking
Woman user (rural or urban)	Installs via WhatsApp shared link, audio-first onboarding	Listens to audio stories in Twi or Pidgin, answers questions, follows the women's livelihood and enterprise track	Practical climate knowledge relevant to her daily life, enterprise opportunities, safety information
Person with disability	Installs via any browser, screen reader or high-contrast mode activates automatically	Navigates fully accessible interface, follows disability track, accesses sign language video options	Accessible climate education, disability-specific preparedness information, accessible green job pathways
Youth (job-seeker or student)	Installs via campus link or WhatsApp	Follows youth and green jobs track, uses career pathway explorer, completes fact-check challenges	Green career guidance, skills passport, CV builder, entrepreneurship toolkit
Institutional partner (NADMO, EPA, NGO)	Accesses partner dashboard via separate web login, no app required	Views BIPI Pulse heat maps, exports aggregated regional data, monitors knowledge gap trends	Real-time national climate knowledge intelligence, demographic breakdown, district-level vulnerability signals
Platform team	Admin dashboard in Supabase	Reviews content performance, flags weak modules, monitors data quality, manages user reports	Drop-off rates, format performance data, data quality signals from Verified Reporter network

12. User Testing Plan

The bootcamp session on UX/UI, Inclusive Design, and User Testing, led by Dr Sarah Dsane, made clear that mentors will assess not just whether a platform claims to be inclusive but whether the team has validated that claim with real users. This is the user testing plan for BIPI's three-month build phase.

Phase	Participants	What is being tested	When	Output
Internal pilot	5 team members plus 10 classmates	Onboarding flow, content format preferences, quiz engine accuracy, offline caching reliability, accessibility features at basic level	Month 1, Week 2	Bug report, onboarding friction log, initial preference data
Extended pilot	30 users recruited from UG and partner schools	Full learning experience across all five formats, gamification loops, BIPI Pulse data quality, cross-device progress sync for Verified Profile users	Month 1, Week 4	Format preference breakdown, completion rate by module, first real BIPI Pulse data
Demographic testing	5 women users (2 rural, 3 urban), 5 youth users, 3 users with disabilities (mixed)	Twi and Pidgin interface usability, audio story format, screen reader compatibility, sign language video, onboarding accessibility, physical interaction with mini games	Month 2, Week 2	Accessibility audit findings, format failure points, demographic completion rates

Institutional testing	2 to 3 potential institutional partners (NGO staff or government agency staff)	BIPI Pulse dashboard usability, data quality, partner-facing analytics value, export functionality	Month 3, Week 1	Dashboard iteration notes, institutional partnership interest signals

All testing findings are documented and presented at each mentor review stage. The team treats user testing not as a box to check but as the primary mechanism for improving the platform between reviews.

---

13. How BIPI Meets Every Assessment Criterion

The proposed schedule confirms that submissions will be scored by mentors using an agreed assessment criteria and scoring framework. Based on the bootcamp structure, those criteria map directly to the five thematic sessions: climate change, software engineering, persuasive technology, accessibility and personalisation, and UX/UI and inclusive design. Here is how BIPI performs against each.

Assessment criterion	How BIPI meets it	Evidence
Climate change relevance and impact	Content is built from the GreenRes curriculum matrix, module by module. All 120 modules are accounted for across five tracks. Content is grounded in Ghana-specific climate events and hazards. BIPI Pulse generates a live national knowledge gap map that is a direct, measurable climate impact beyond content delivery.	Five learning tracks covering all 120 modules. BIPI Pulse data layer. Regional heat maps. Module 97 (Citizen Science) direct alignment.
Innovation	No learning platform in Ghana passively generates a national citizen science climate knowledge map. This is a genuinely new category, not an incremental improvement on existing tools. The two-tier identity system, the passive data layer, and the institutional dashboard together create something that does not currently exist in the Ghanaian edtech or climate education landscape.	BIPI Pulse architecture. Module 97 validation. Institutional dashboard. Citizen science framing.
Technical feasibility	The stack is proven and the scope is realistic. Next.js 15 and Supabase are used by student teams globally. The team has four Computer Science students who can build and deploy a full-stack PWA. Five demo modules with a live BIPI Pulse dashboard is a credible, achievable target for the first review. Full offline capability is built in from the architecture level.	Tech stack (Section 10). Three-month roadmap (Section 14). Offline-first design. Claude API fallback plan (Section 11).
Inclusion and accessibility	Accessibility is the default, not an option. The platform serves three marginalised groups with dedicated content tracks and format adaptations. Screen reader support, high contrast, sign language video, TTS, Twi and Pidgin, offline mode, and	Section 6 (Inclusion in detail). Disability track (modules 50 to 57). Women's track (modules 42 to 49). Audio story format. Accessibility defaults.

	SMS fallback are all built into the base product. BIPI Pulse tracks inclusion outcomes quantitatively.	
Persuasive technology	Progress bars, XP points, district badges, streak counters, district leaderboard, social proof via district competition, loss aversion via streak loss, personal action tracker via reflection journal, and the Verified Reporter badge as an accuracy incentive. All mechanisms are named and grounded in the session content from Dr Akon Obu Ekpezu and Dr Prince Boakye-Sekyerehene.	Section 4 (content formats). Gamification design. Reflection journal. BIPI Pulse leaderboard. Persuasive technology session alignment.
UX/UI and inclusive design	Five content formats matched to five distinct user need types. Zero login wall. Three-tap onboarding with no keyboard required. Personalisation from the first interaction. Four-phase user testing plan with demographic-specific testing for women, youth, and persons with disabilities. Dr Sarah Dsane's session criteria are directly addressed.	Section 8 (user pipeline). Section 12 (user testing plan). Onboarding design. Personalisation logic.
Team capability and collaboration	Four Computer Science students covering frontend, backend, AI and content engineering. One Law student owning Ghana Data Protection Act compliance, the consent framework, and institutional partnership language. Every team member has a distinct, non-overlapping role. The Law student's contribution is unique and directly addresses a gap that almost every other team will ignore.	Section 13 (team). Consent framework. Ghana Data Protection Act 2012 (Act 843) compliance. Role distribution.

14. Team

The team is five people. Four are Computer Science students. One is a Law student. The combination is intentional. The technical team can build the platform. The Law student makes it trustworthy and legally defensible, which matters enormously for a platform asking users to consent to data collection.

Role	Background	What they build on BIPI
Frontend and PWA Lead	Computer Science	The entire user-facing application: app shell, service worker for offline caching, PWA install flow, all five content format interfaces, the gamification UI including progress bars, XP display, badge system, streak counter, and district leaderboard. Also responsible for Twi and Pidgin language switching at the UI level and text-to-speech integration.
Backend Engineer	Computer Science	Supabase schema design and management, all API routes between the frontend and the database, the device UUID identity system, the BIPI Pulse aggregation pipeline that converts raw response data into anonymised district-level intelligence, and the partner-

		facing dashboard backend.
AI and Content Engineer	Computer Science	Claude API integration for module content generation and Twi/Pidgin localisation, the quiz engine logic, the AI career advisor feature in the youth track, the offline fallback content generation and caching strategy, and quality review of all AI-generated module content against the curriculum matrix.
Design and Product Lead	Computer Science	UX design for all screens, the onboarding flow, the Ghana map interface for regional features, the institutional BIPI Pulse dashboard visualisations, the demo narrative for mentor reviews and the final pitch, and all impact metrics documentation.
Data Governance Lead	Law	Ghana Data Protection Act 2012 (Act 843) compliance review, the consent screen text in Twi, Pidgin, and English, the data ethics framework for BIPI Pulse, institutional partnership agreement language, and legal review of any data-sharing arrangements with government agencies or NGOs.

17. Sustainability: How BIPI Keeps Running

BIPI is designed to be sustainable because it becomes more valuable over time as more users join and BIPI Pulse gets richer. The platform that starts with fifty users generating thin data becomes dramatically more useful with five thousand users generating district-level climate intelligence. The data is the moat.

Revenue stream	How it works	Timeline
Institutional dashboard subscriptions	NADMO, EPA Ghana, MoFA, NGOs, and development organisations pay a subscription fee to access the BIPI Pulse partner dashboard with full regional analytics, demographic breakdowns, and data export.	From month 4, after first real data is available
Development partner licensing	International development organisations including GIZ, USAID, and UN agencies license BIPI to deploy targeted climate education programmes in their priority communities, using BIPI's existing infrastructure and content.	From month 6, with first institutional test results as evidence
Government education contracts	Ghana Education Service and district education offices embed BIPI modules into school climate curricula as a certified digital learning tool, with BIPI providing the platform and content.	Year 2, after platform is demonstrated at scale
Grant continuation	The initial Mastercard Foundation and ACC hackathon grant is the foundation. BIPI Pulse data provides the evidence base for multi-year climate education grant applications, because the platform can show exactly where funding is needed and what impact previous funding achieved.	Ongoing from month 3
Freemium model	Core learning is always free for individual users. Verified profile, downloadable certificates, extended leaderboard, and premium module unlocks are a small-fee tier	From month 4

	for organisations and highly motivated individual learners.	

A platform that turns every Ghanaian who learns about climate change into a data point that makes Ghana smarter about climate change.

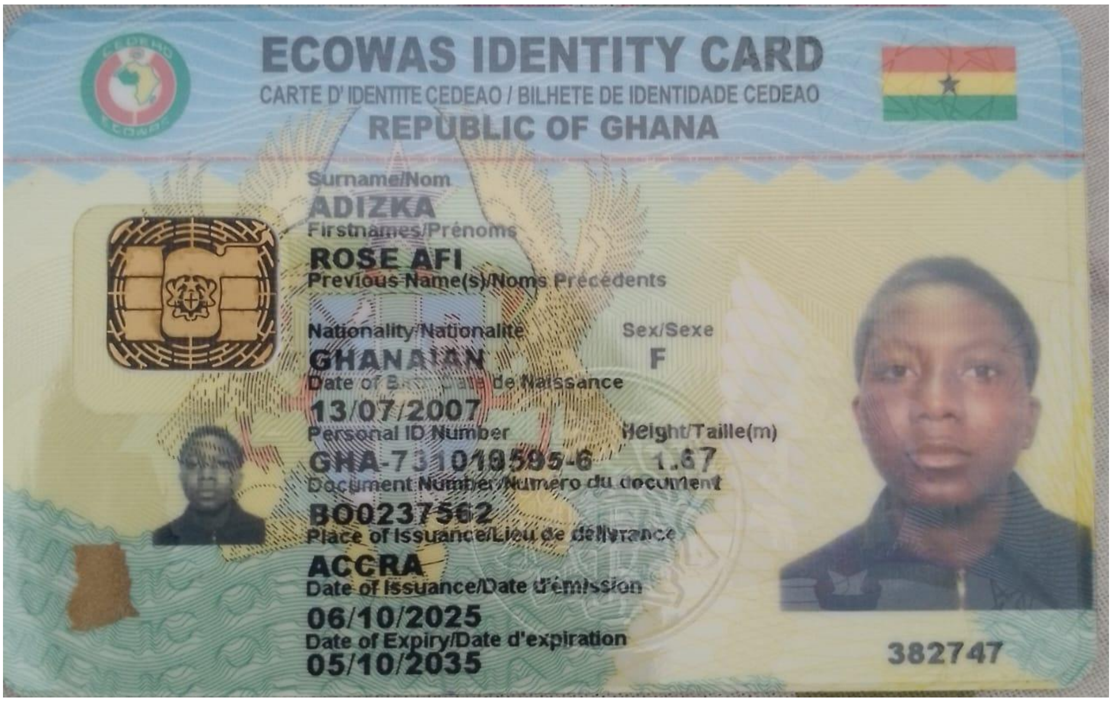
The GreenRes curriculum matrix told this team exactly what to build. Every module has a content format, an assessment method, and a suggested platform feature. The team read all 120 of them and built a system that delivers every format, every assessment method, and every suggested feature. Not approximately. Exactly.

The judges wrote Module 97 themselves. It is called Citizen Science. It calls for a citizen science dashboard, a reporting map, a community leaderboard, and a mobile reporting interface. BIPI Pulse is the fulfilment of that module. When the judges see it, they will recognise their own vision built back at them, with a live national heat map running on top of it.

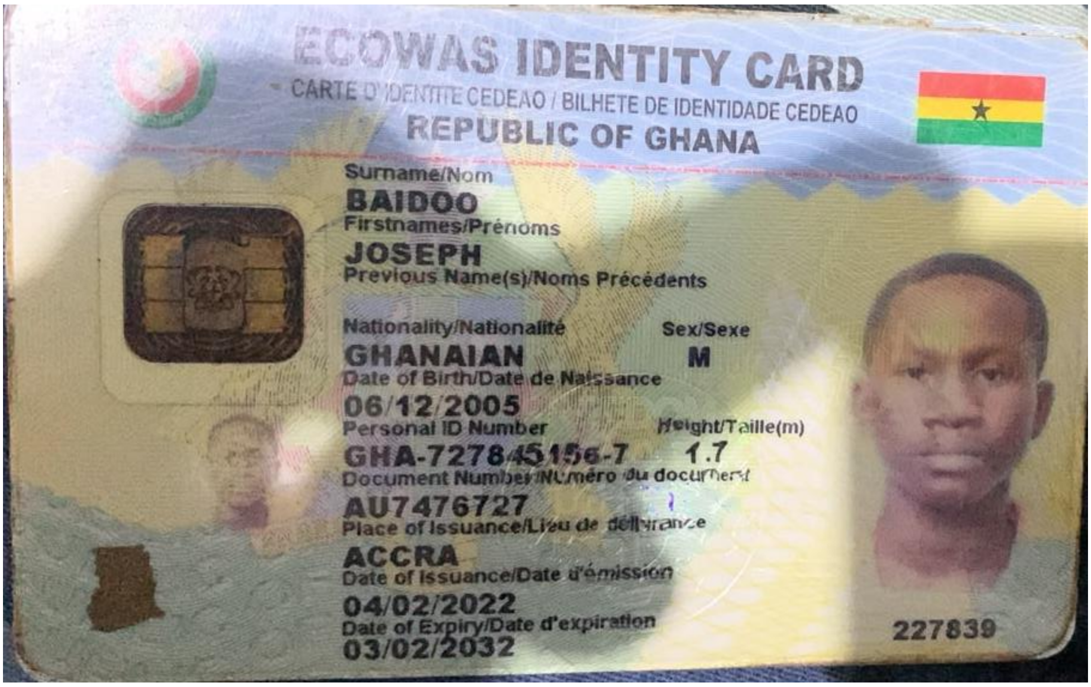
The learning platform is the product. The national knowledge map is the public good. The blood pressure reading is what Ghana needs before the next crisis hits.

BIPI: Ghana's Ecological Pulse. Know the Signs. Act Before the Crisis.

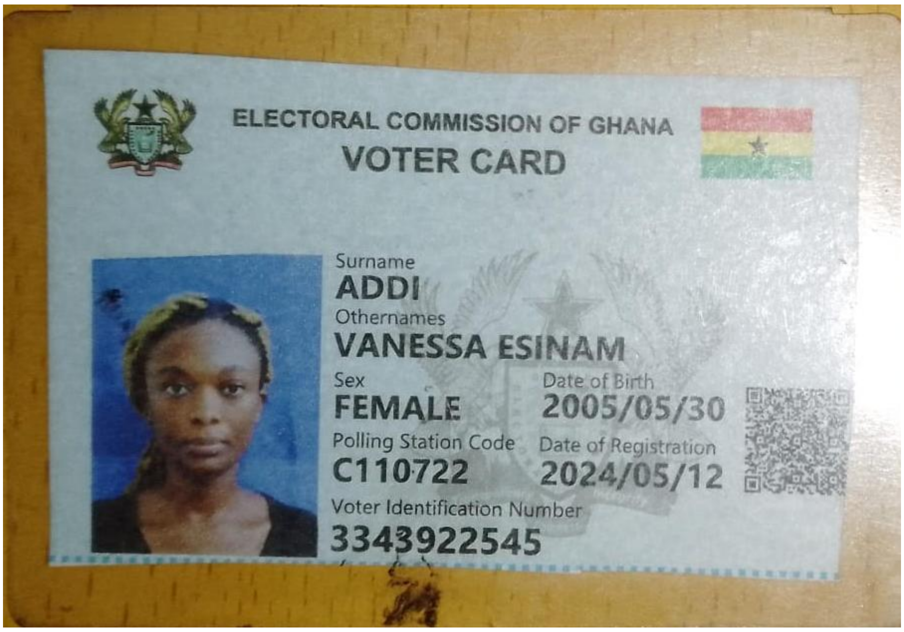
Sonpon Ye-shua Chief Ghana card (front and back)



Rose Afi Adzika Ghana card (front and back)



Baidoo Joseph Ghana Card (front and back)



Vanessa Addi Essinam (Voter's ID)

Patience Nana Ekua Asantewaa Ghana card (fornt and back)